

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer from all the Sections as directed.

Section – A

(Objective Type Questions)

(Compulsory)

1. Select the most suitable answer from those given
or Fill in the blanks : $1 \times 10 = 10$

(a) Pollutants that are easily decomposable in
nature are called :

- (i) Biodegradable pollutants
- (ii) Non-biodegradable pollutants
- (iii) Renewable pollutants
- (iv) None of these

(b) The main components of photochemical smog is :

- (i) Water vapour (ii) Nitrogen oxides
- (iii) Sulphur oxides (iv) All of these

(c) Aerosols consisting of solid particles produced by combustion is :

- (i) Fog (ii) Smog
- (iii) Smoke (iv) None of these

(d) Which of the following material can be recycled multiple times ?

- (i) Aluminium
- (ii) Wood
- (iii) Organic materials
- (iv) Plastic

(e) Which gas is generated on the decomposition of the organic matter present in the sanitary landfill ?

- (i) Oxygen (ii) Methane
- (iii) Nitrogen (iv) Hydrogen

(f) Solid waste burning in presence of oxygen is known as _____.

- (g) Landfill are the large open areas used to dispose solid waste.
- (h) The first step in sewage treatment is _____.
- (i) The state of increased levels of Nitrogen and Phosphorous in the lake is called Eutrophication.
- (j) The process of decomposition of biodegradable solid waste by earthworms is called _____.

Section – B

(Short-answer Type Questions)

3. Write short notes on any four of the following :

5×4 = 20

- (a) Sewage treatment plant
- (b) Characterization of municipal solid waste
- (c) Impact of solid waste on plant health
- (d) Thermal treatment of waste material
- (e) Biodegradable and Non-biodegradable waste
- (f) E-Waste

Section – C

(Long-answer Type Questions)

Answer any four questions of the following :

10×4 = 40

3. Write explanatory notes on Solid Waste Management.
4. What is Bio-medical waste ? Write a short account of Bio-medical Waste Management.
5. Discuss the importance of effluent treatment plant in controlling water quality.
6. What are the chief sources of solid waste ? Write a short account of harmful effect of metal pollutants on human health.
7. Describe the effect of industrial waste on air, water and soil.
8. Discuss about the classification and chemical composition of solid waste.

